Anshu Nigam

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Education

Institute of Engineering and Management, Kolkata **2024 – 2028**
B.Tech in Computer Science and Engineering (CGPA: 8.36) *Kolkata, India*

Relevant Coursework

- Data Structures & Algorithms
- Probability & Statistics
- Linear Algebra
- Intro to Machine Learning

Publications

HARD-RL: Reinforcement Learning for Fog Computing Load Balancing– *IJACSA, submission pending publication (2026).* Extended Q-learning-based fog load balancing with a **Fuzzy Logic Gate, adaptive reward mechanism, and energy-aware ϵ-decay**, achieving a **98.29% reduction in failed task allocations** and **32.91% lower per-task energy consumption**.

Projects

Credit Card Fraud Detection System | *Random Forest, SHAP, FastAPI* **2026**
- Engineered a fraud detection model (Random Forest) achieving 0.99 AUC on a highly imbalanced dataset, layering SHAP-based explainability for interpretable predictions.
- Deployed the model via a FastAPI backend with an animated real-time HTML dashboard for live inference.

AutoBench — No-Code AutoML Platform | *FastAPI, Next.js 14, TypeScript* **2026**
- Architected a full-stack, no-code AutoML SaaS platform enabling users to train and benchmark models across 12 ML algorithms without writing code.
- Built Models, Datasets, and Experiments dashboards with real-time KPI stat cards and podium-style leaderboards, abstracting the full ML pipeline from preprocessing to evaluation.

AI-Based Resume Parser & Candidate Ranking System | *Python, NLTK, Scikit-learn* **2026**
- Built an NLP pipeline (tokenization, stemming, lemmatization, TF-IDF + cosine similarity) layered with K-Means, KNN, and Decision Tree classifiers to rank candidates into a 0–10 job-fit score.

Handwritten Digit Recognizer | *Convolutional Neural Network (CNN)* **2026**
- Trained a CNN achieving 98.93% test accuracy on handwritten digit classification and deployed a live interactive demo on Hugging Face Spaces.

Telecom Customer Churn Prediction | *Logistic Regression, Random Forest, XGBoost, SMOTE* **2026**
- Benchmarked Logistic Regression, Random Forest, and XGBoost churn models, applying SMOTE to correct class imbalance and improve recall on the minority churn class.

Email Spam Detection | *SVM, TF-IDF, NLP* **2026**
- Developed an email spam classifier using SVM with TF-IDF vectorization and NLP preprocessing; documented in a Jupyter notebook and formal report.

Technical Skills

Languages: Python, Java, C
ML / DL: Scikit-learn, XGBoost, SHAP, CNNs, NLTK, TF-IDF, Q-Learning / Reinforcement Learning
Data Analysis: Pandas, NumPy, feature engineering, EDA
Techniques: NLP preprocessing, K-Means, KNN, Decision Trees, SMOTE, cosine similarity, from-scratch algorithm implementation
Tools & Deployment: FastAPI, Jupyter, Git/GitHub, Hugging Face Spaces
Math Foundations: Probability & Statistics, Linear Algebra

Certifications

Apna College — Sigma 8.0 Batch (Completed) | CodeWithHarry — Data Science Course (Completed)

Achievements

- Self-directed ML curriculum implementing ensemble methods from scratch (Naive Bayes through Gradient Boosting, XGBoost, and Stacking), structured around an 8-phase AI/ML roadmap from math foundations through generative AI/LLMs.
- Maintains a public GitHub portfolio spanning NLP, computer vision, and reinforcement learning projects, from research to deployed inference demos.